

SOFTWARE DESIGN & UML ARCHITECTURE SPECIFICATION

For PharmaCart - Omni-Channel Pharmacy Ordering Platform

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Document Control & Metadata

- Project Name:** PharmaCart — Online Ordering for Neighbourhood Pharmacy Chain
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- Author / Lead Software Architect:** Ritesh Jadhav
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Revision History

Version	Date	Author	Description of Change	Approved By
0.1	2026-10-02	Ritesh Jadhav	Domain modeling and actor-usecase mapping	Tech Lead & Architecture Review
0.8	2026-10-03	Ritesh Jadhav	Sequence, activity, and state machine models drafted	Lead Systems Architect
1.0	2026-10-05	Ritesh Jadhav	Finalized UML package, consistency matrix, coupling report	Engineering Steering Committee

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1. Architectural Overview & Design Principles

The architectural design of PharmaCart balances high statutory compliance (Drugs & Cosmetics Act) with the operational realities of 14 disconnected legacy POS systems across Ahmedabad.

Architectural Tier	Component Layer	Technologies & Protocols	Primary System Function
Presentation Tier	Customer Web/Mobile SPA, Pharmacist Desk, Store Counter POS	HTML5, React, Tailwind, WebSockets	End-user interfaces for order placement, clinical verification, and store fulfillment.
Security & Gateway	API Gateway, Rate Limiter, JWT Auth Interceptor	Nginx / Kong, TLS 1.3, HTTPS (443)	Enforces authentication, request throttling, and OWASP security controls.
Domain Core Tier	Order Service, Prescription Service, Store Routing Engine	Node.js / Python FastAPIs, Event Bus	Core business logic, clinical review queues, and multi-store reservation orchestration.
Infrastructure Tier	PostgreSQL 16 DB, S3 Prescription Vault, 14 POS Daemons	AES-256 Storage, mTLS, Background Daemons	Relational transactional persistence, encrypted medical file storage, and POS synchronization.

The system is governed by the following core design principles:

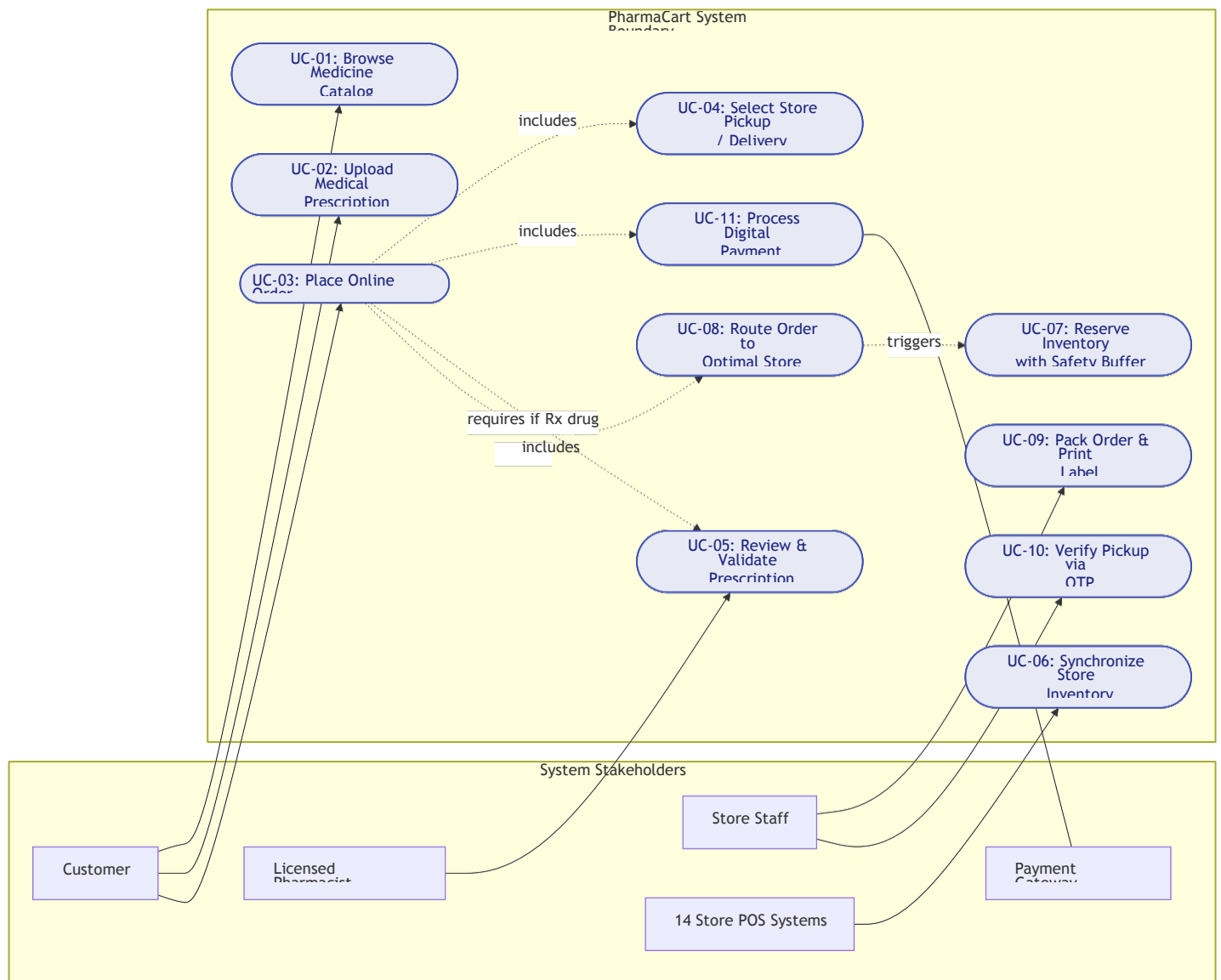
- Separation of Concerns (SoC):** Clinical compliance logic (prescription inspection) is isolated from commercial fulfillment logic (inventory reservation and payment processing).
- Event-Driven Asynchrony:** Communication between long-running human workflows (pharmacist manual inspection) and system transactions (payment capture) relies on event streaming rather than blocking synchronous threads.
- Fail-Safe Inventory Degradation:** The inventory architecture operates under the assumption that local POS sync may fail or experience delays; the **Safety Stock Buffer** (Available = Physical - 2) acts as an automated safety cushion.

2. Use Case Analysis & Diagram

2.1 Actor Catalog

Actor Name	Type	Operational Responsibility in PharmaCart
Customer	Primary (Human)	Browses catalog, uploads prescriptions, places delivery/pickup orders, tracks order progress, provides pickup OTP.
Licensed Pharmacist	Primary (Human)	Inspects prescription documents, validates doctor credentials, approves or rejects medication requests, issues clinical modifications.
Store Fulfillment Staff	Secondary (Human)	Receives allocated picking slips, picks physical medications from store racks, verifies barcodes, validates pickup OTP.
Delivery Runner	Secondary (Human)	Picks up packaged orders from store, executes doorstep hand-over, collects cash (if COD), collects delivery signature/OTP.
Legacy Store POS System	Secondary (System)	Automated client running at each of the 14 stores; transmits delta inventory snapshots and receives temporary reservation locks.
Payment Gateway	Supporting (System)	Authorizes credit/debit card, net banking, and UPI transactions; delivers asynchronous webhook event callbacks.
SMS/Notification Gateway	Supporting (System)	Dispatches transactional SMS, WhatsApp messages, order status updates, and verification OTPs.

2.2 Use Case Diagram



2.3 Detailed Use Case Specifications

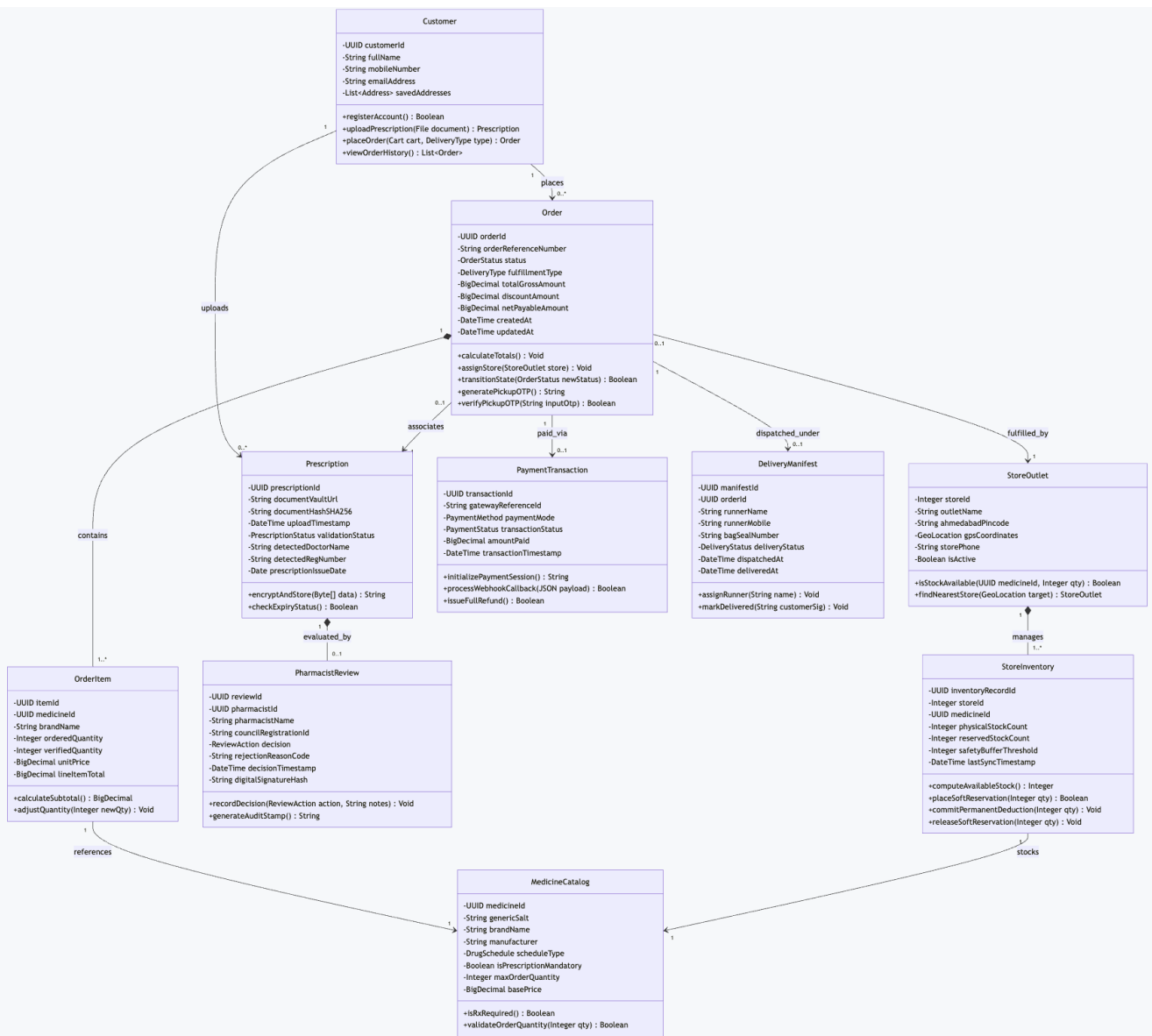
Use Case: UC-05 — Review & Validate Medical Prescription

Field	Detail
Use Case ID	UC-05
Use Case Name	Review & Validate Medical Prescription
Primary Actor	Licensed Pharmacist
Trigger	An order containing one or more Schedule H/H1 /X medicines is placed by a customer.
Preconditions	1. The customer has uploaded a readable image/PDF prescription. 2. Pharmacists are authenticated on the Tele-Pharmacy Portal with active council registration.
Main Success Scenario	1. Pharmacist opens the next pending item from the central FIFO queue. 2. The system displays high-res prescription documents alongside mapped order items. 3. Pharmacist verifies doctor' s name, council registration number, clinic stamp, patient name, and prescription date (< 30 days). 4. Pharmacists confirms prescribed drugs match ordered items, strengths, and maximum dosages. 5. Pharmacists clicks Approve and enters their 4-digit digital signing PIN. 6. System logs approval with timestamp and council registration ID. 7. System transitions order state to APPROVED_AWAITING_ALLOCATION .
Alternative Flows	4a. Prescription Illegible / Expired / Forged: 1. Pharmacist clicks Reject and selects reason (e.g., "Doctor seal missing", "Prescription older than 30 days"). 2. System transitions order to RX_REJECTED . 3. Notification Gateway sends SMS/WhatsApp to customer explaining rejection with re-upload link. 4b. Partial Dosage Available: 1. Pharmacists adjusts prescribed quantity downward (e.g., from 30 tablets to 1 0). 2. System updates order subtotal and alerts customer for re-confirmation.
Postconditions	Digital audit log permanently recorded. Order unblocked for store inventory reservation.

3. Domain Class Architecture & Diagram

3.1 Class Diagram

The class diagram below depicts the structural domain model of PharmaCart. Notice the deliberate decoupling between Prescription / PharmacistReview and StoreInventory / StoreOutlet . The central Order entity orchestrates relationships through clean association interfaces.



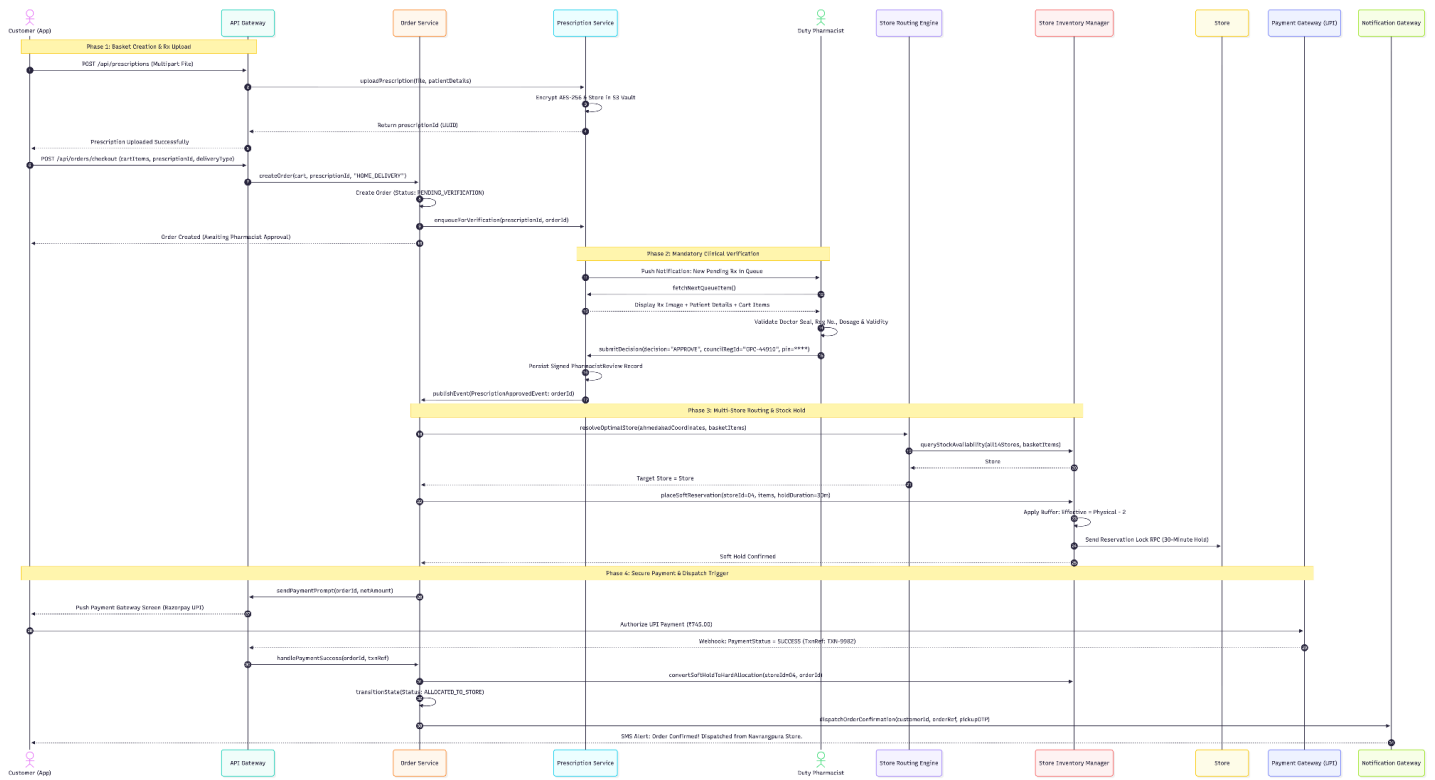
3.2 Class Dictionary & Responsibility Matrix

Class Identifier	Core Architectural Responsibility	Cohesion Rating	Primary Collaborators
Customer	Manages user identity, stored delivery locations in Ahmedabad, and patient profile attributes.	High (Identity)	Order , Prescription
Order	Aggregate Root managing checkout lifecycle, line item computations, status changes, and OTP generation.	High (Transaction)	OrderItem , StoreOutlet , PaymentTransaction
Prescription	Encapsulates prescription metadata, AES-256 vault URI, cryptographic document integrity, and expiry checks.	High (Clinical Data)	Customer , PharmacistReview
PharmacistReview	Immutable clinical legal record holding pharmacist identity, council registration number, decision, and digital signature.	High (Compliance)	Prescription
StoreOutlet	Represents physical retail branch in Ahmedabad (e.g., Navrangpura, Satellite), including geo-location and operating state.	High (Physical Entity)	StoreInventory , Order
StoreInventory	Manages store-specific shelf balances, delta updates from POS, soft holds, and safety buffer enforcement.	High (Inventory Math)	StoreOutlet , MedicineCatalog
PaymentTransaction	Handles payment provider session creation, idempotency, webhook processing, and transaction receipts.	High (Financial)	Order
DeliveryManifest	Oversees last-mile logistics dispatch, runner routing, tamper-evident bag seal validation, and doorstep confirmation.	High (Logistics)	Order

4. Interaction Modeling: Sequence Diagram

4.1 Sequence Diagram: 'Upload Prescription & Place Order'

This sequence traces the end-to-end realization of **FR-01, FR-02, FR-05, FR-06, and FR-10**. It highlights the asynchronous inspection gate by the pharmacist and the soft reservation of stock at the optimal retail branch.



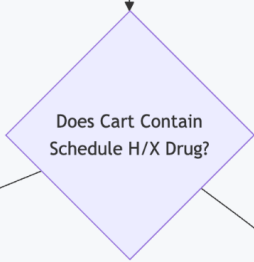
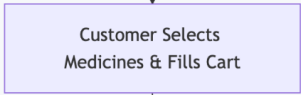
4.2 Message Sequence Trace

Step #	Originator	Target	Message Name / Operation	Semantics
1–3	Customer App	Prescription Service	uploadPrescription()	Synchronous HTTPS multipart upload. Encrypted at rest using AES-256.
4–6	Customer App	Order Service	createOrder()	Synchronous. System records draft order in PENDING_VERIFICATION state.
7–10	Duty Pharmacist	Prescription Service	submitDecision()	Manual human verification. Pharmacist signs record with State Council ID.
11–13	Order Service	Routing Engine	resolveOptimalStore()	Evaluates 14 store geo-locations and inventory balances across Ahmedabad.
14–16	Routing Engine	Store Inventory	placeSoftReservation()	Decrements temporary available units while keeping safety buffer (N - 2) intact.
17–19	Customer App	Payment Gateway	Authorize UPI Payment	Third-party PCI-DSS transaction. Webhook confirms capture asynchronously.
20–22	Order Service	Store Staff & Customer	dispatchOrderConfirmation()	Converts reservation to permanent allocation; triggers store packing slip print.

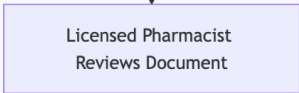
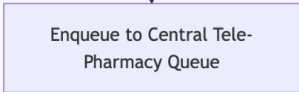
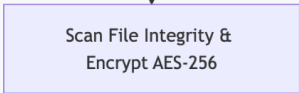
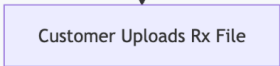
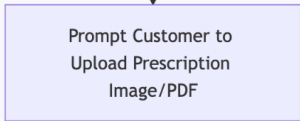
5. Process Modeling: Activity Diagram

5.1 End-to-End Activity Flow

The activity diagram captures decision logic, conditional branches (OTC vs. Rx, in-stock vs. out-of-stock), and parallel processing streams

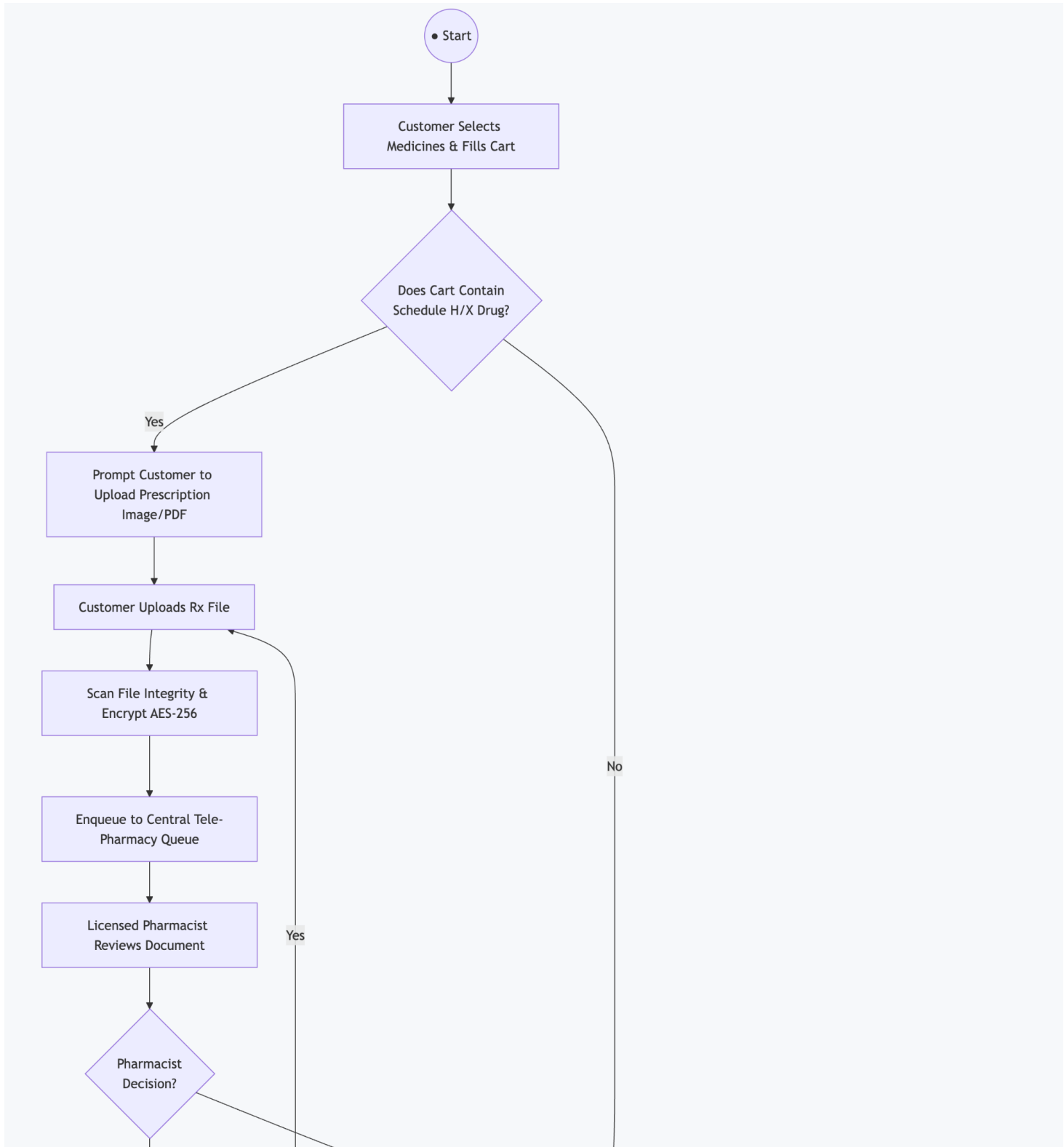


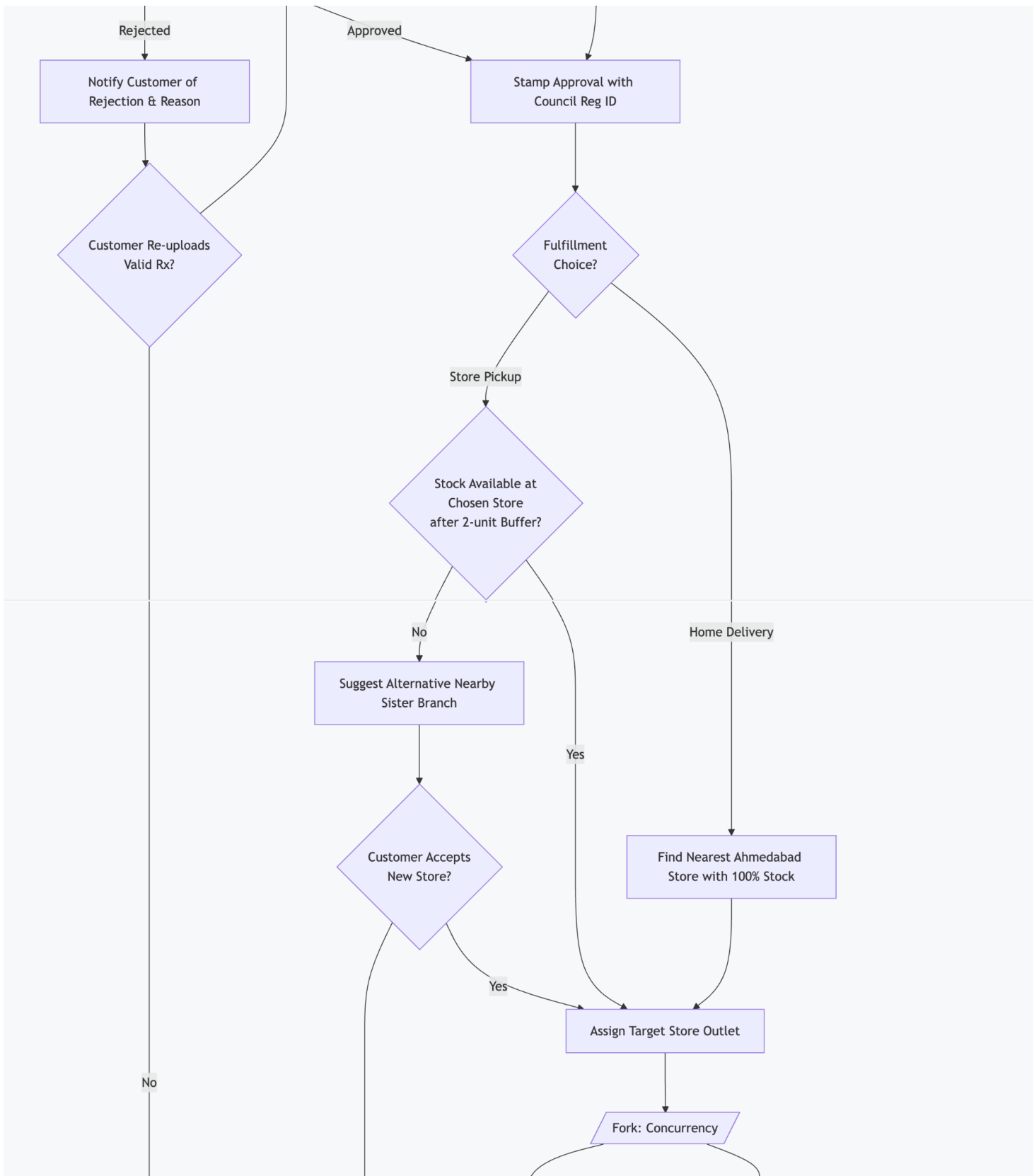
Yes

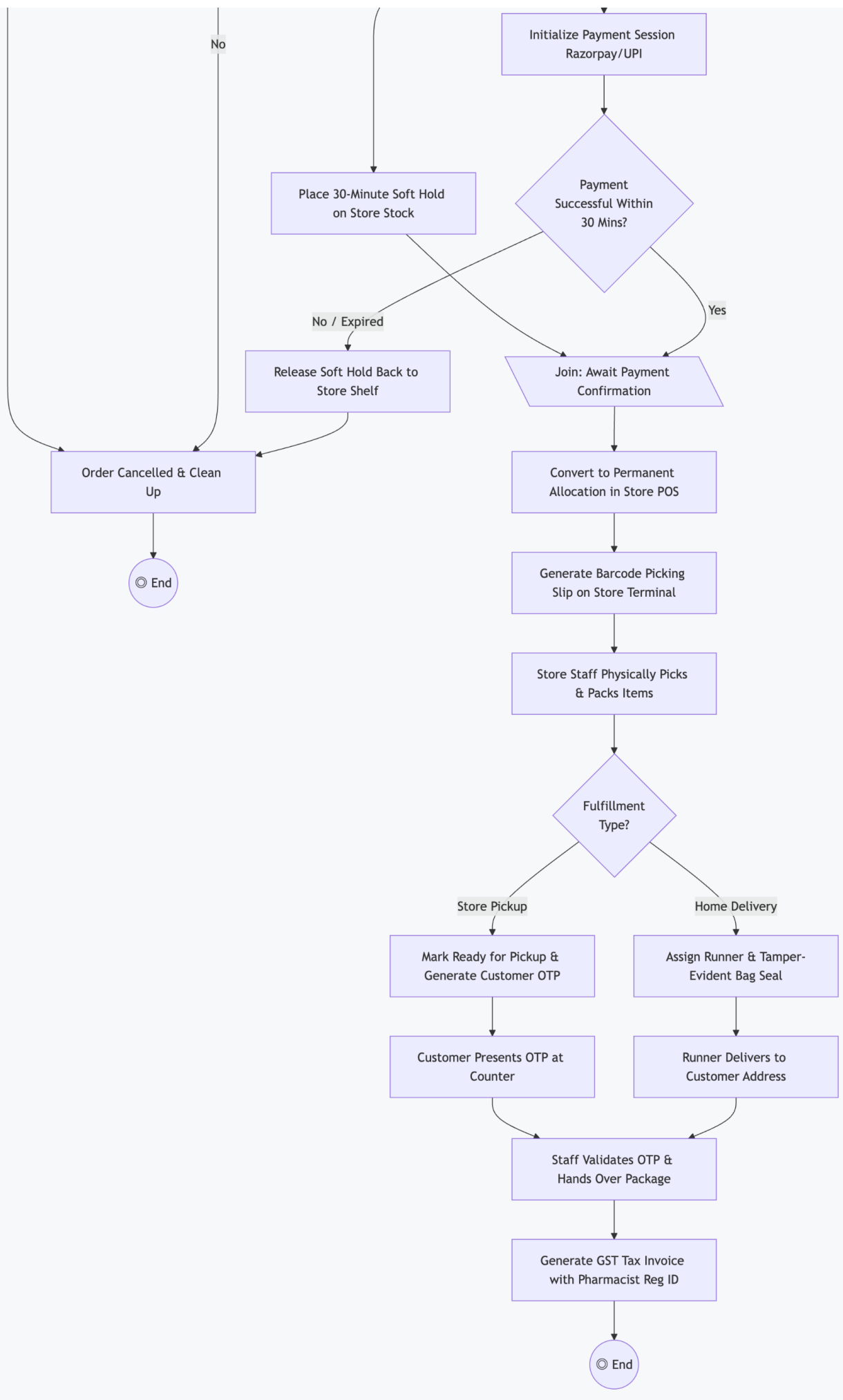


Yes

No







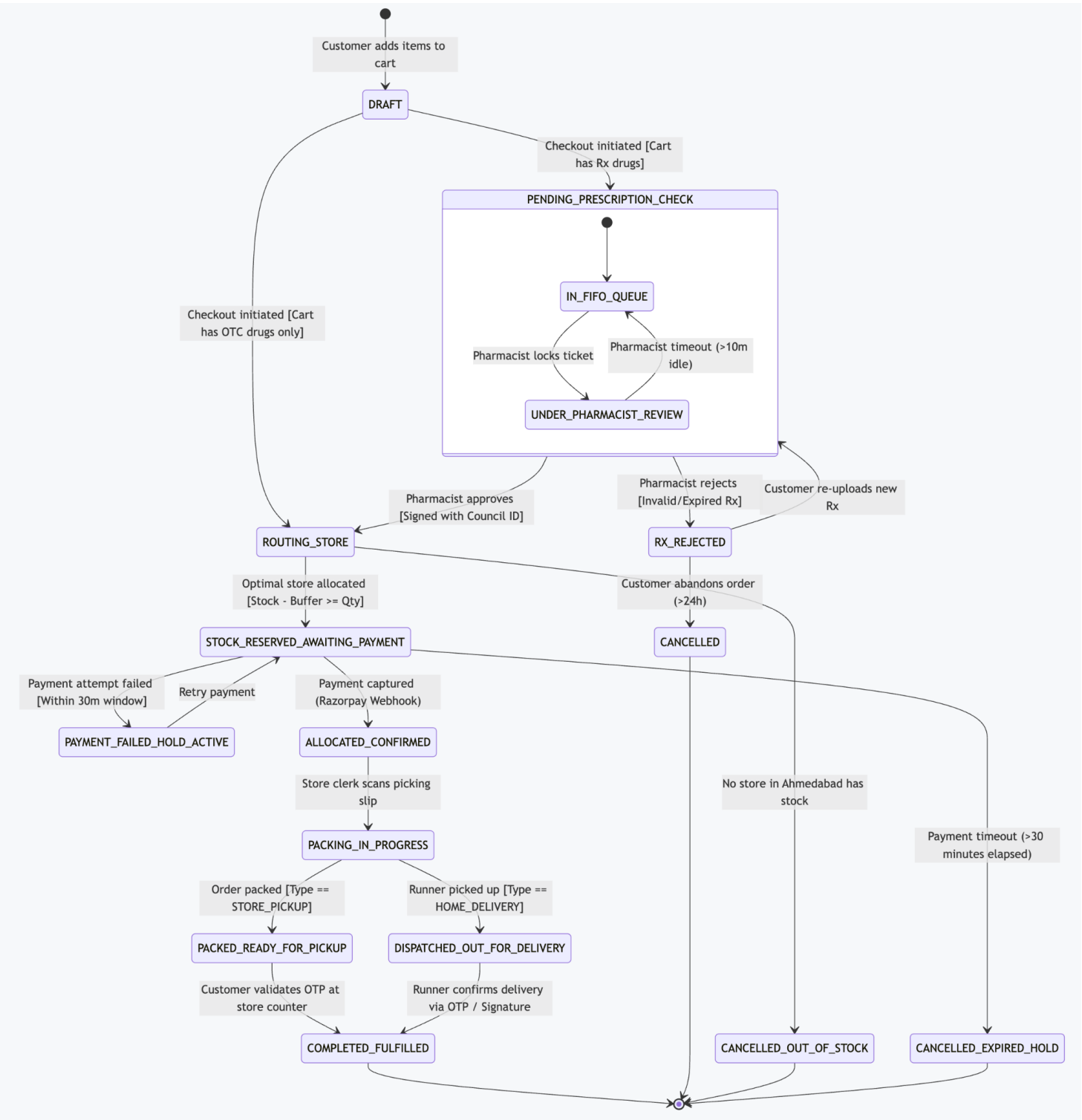
5.2 Decision & Fork-Join Node Specifications

Node Name	Node Type	Input Condition / Invariant	Outgoing Branch Logic
CHECK_RX	Decision Diamond	Cart item collection inspection.	Yes : Enforces upload flow; No : Bypasses to store selection.
DECISION_RX	Decision Diamond	Pharmacist review outcome.	Approved : Moves to routing; Rejected : Alerts customer with reason code.
CHECK_PICKUP_STOCK	Decision Diamond	Available = Physical - 2 >= OrderQty .	Yes : Allocates store; No : Triggers sister-store failover.
FORK_CHECKOUT	Fork Bar	Order allocation completed.	Splits into two concurrent parallel threads: (1) 30-min inventory lock, (2) Payment session.
JOIN_CHECKOUT	Join Bar	Both threads must synchronize.	Synchronizes only if payment succeeds before the 30-minute lock expires.

6. Behavioral Modeling: State Machine Diagram

6.1 State Machine Diagram for 'Order' Lifecycle

The order entity serves as the stateful transactional heart of PharmaCart. The diagram below illustrates every valid operational state, transition event, and exception guard.



6.2 State Transition Table

Source State	Event Trigger	Guard Condition	Target State	Action / Side Effect Executed
DRAFT	INITIATE_CHECKOUT	Cart contains ≥ 1 Schedule H/X drug	PENDING_PRESCRIPTION_CHECK	Generate Prescription_ID , lock file in S3 vault.
DRAFT	INITIATE_CHECKOUT	Cart contains OTC items only	ROUTING_STORE	Bypass clinical verification; query store stock.
UNDER_REVIEW	SUBMIT_DECISION	Action == APPROVE	ROUTING_STORE	Record pharmacist council registration number and timestamp.
UNDER_REVIEW	SUBMIT_DECISION	Action == REJECT	RX_REJECTED	Dispatch SMS to customer with specific rejection reason.
ROUTING_STORE	STORE_FOUND	Store physical stock - 2 ≥ order qty	STOCK_RESERVED_AWAITING_PAYMENT	Place 30-min soft lock on store POS inventory.
STOCK_RESERVED	HOLD_EXPIRED	Time elapsed since hold > 30 minutes	CANCELLED_EXPIRED_HOLD	Release soft lock; restore stock availability badge.
STOCK_RESERVED	PAYMENT_SUCCESS	Webhook signature verified	ALLOCATED_CONFIRMED	Commit permanent inventory deduction; generate picking slip.
PACKING	BAG_SEALED	Order fulfillment == STORE_PICKUP	PACKED_READY_FOR_PICKUP	Generate 6-digit cryptographic OTP; SMS to customer.
PACKED_READY	VERIFY_OTP	Input OTP matches hashed store OTP	COMPLETED_FULFILLED	Mark transaction complete; issue GST e-invoice.

7. Architectural Justification: Coupling & Cohesion Analysis

7.1 Loose Coupling: Prescription Verification vs. Store Inventory

A critical architectural mandate of Case Study No. 68 is explaining **how Prescription Verification and Store Stock are kept loosely coupled**.

In naive architectures, the prescription verification module calls the inventory database directly to check stock, or the inventory module blocks orders while awaiting pharmacist review. This tight coupling creates catastrophic operational vulnerabilities:

- If the pharmacy billing software at Store #04 crashes or loses internet connectivity, the pharmacist cannot review prescriptions.
- If the pharmacist verification portal suffers high latency, inventory reservation locks expire prematurely, generating phantom stockouts.
- Medical information (doctor notes, patient diagnoses) leaks into the commercial store billing database, violating the Digital Personal Data Protection (DPDP) Act.

Our Decoupling Mechanism: The Asynchronous Domain Event Boundary

Architecture Dimension	Clinical Verification Context	Commercial Store Inventory Context
Domain Scope	Prescription validity, doctor council ID, dosage checks	1 4-store shelf stocks, POS delta sync, rack locations
Data Ingestion	Patient uploads via App / S3 encrypted vault	On-premise POS Daemons via outbound HTTPS / WSS
Processing Cadence	Asynchronous human verification (Minutes to Hours)	Near-real-time automated reservation (Milliseconds)
Cross-Boundary Coupling	Emits lightweight PrescriptionApprovedEvent only	Listens for approved event to trigger reservation
Privacy Isolation	Encrypted medical images, doctor notes, patient PHI	Zero clinical data; receives only approved SKU IDs & counts

Clinical Bounded Context

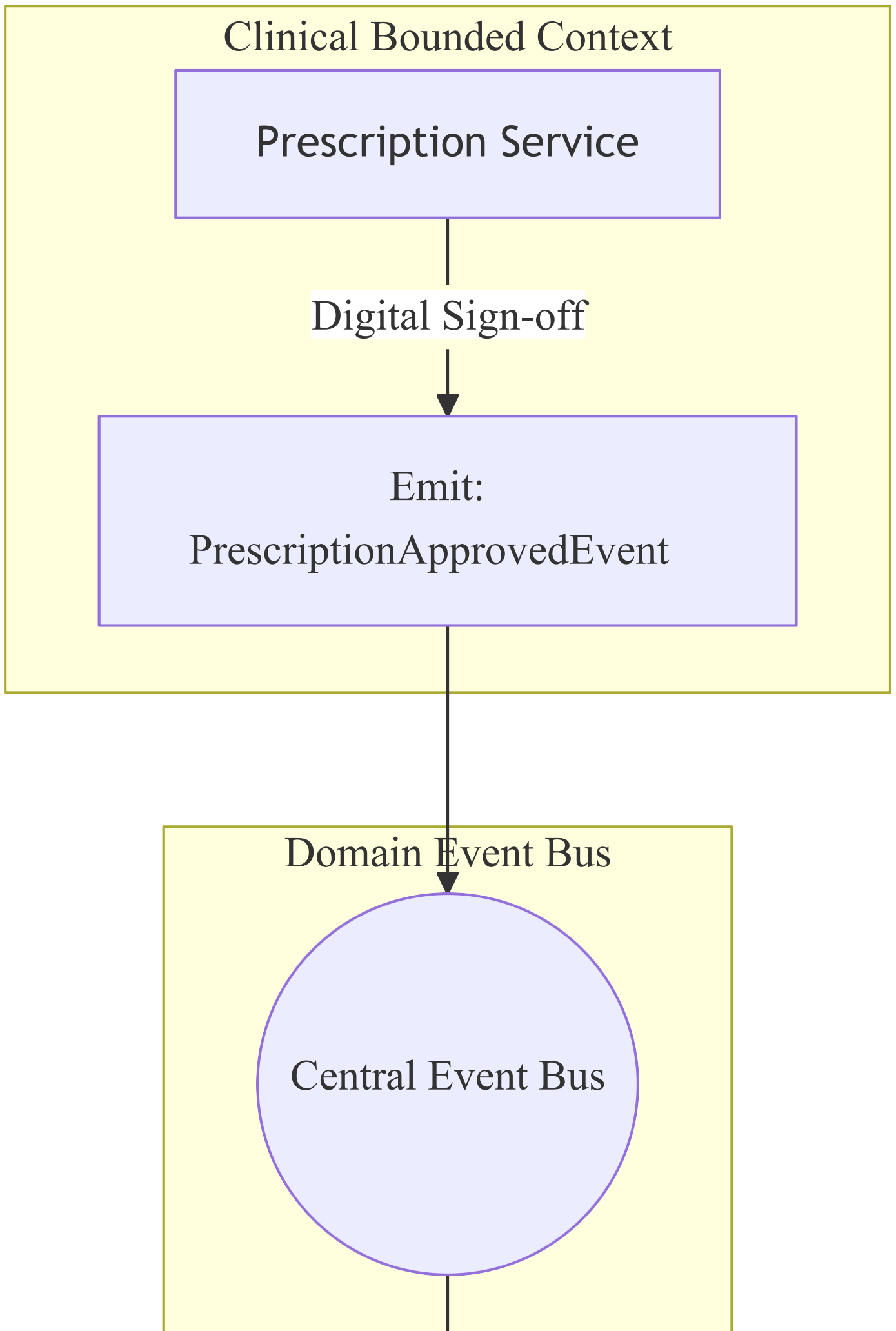
Prescription Service

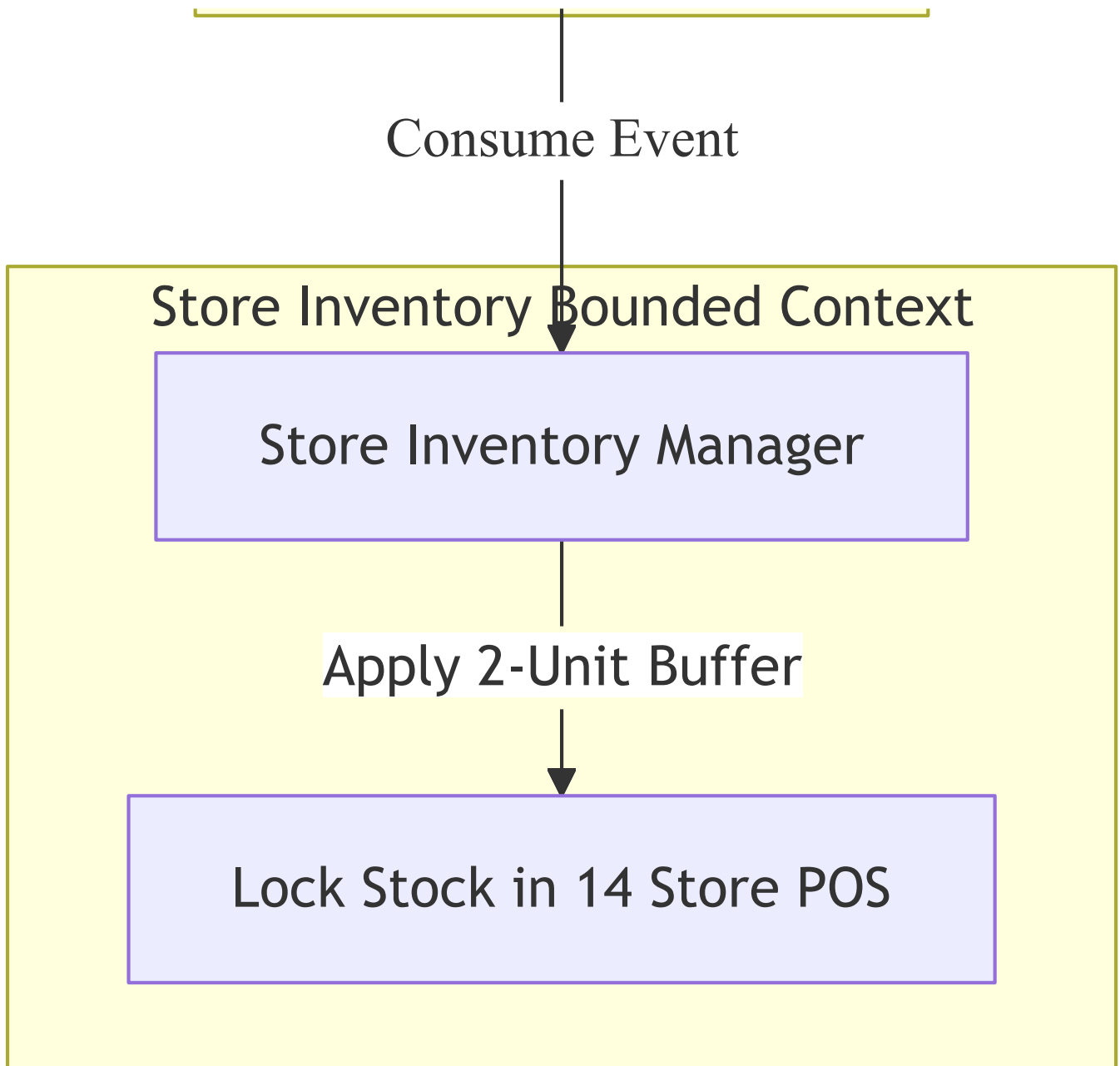
Digital Sign-off

Emit:
PrescriptionApprovedEvent

Domain Event Bus

Central Event Bus





1. **Information Hiding:** The PrescriptionService knows nothing about store locations, shelf quantities, or POS billing software formats. It solely validates clinical legality and emits a clean domain event: PrescriptionApprovedEvent .
2. **Schema Independence:** The store inventory schema (Store_ID , Rack_Location , Physical_Count , Safety_Threshold) is completely quarantined from the clinical verification schema (Doctor_Reg_Number , Clinic_Seal_Valid , Pharmacist_License_ID) .
3. **Temporal Decoupling:** Verification operates at human speed (minutes to hours), while inventory reservation operates at machine speed (milliseconds). The order entity transitions between them asynchronously without holding open database locks.

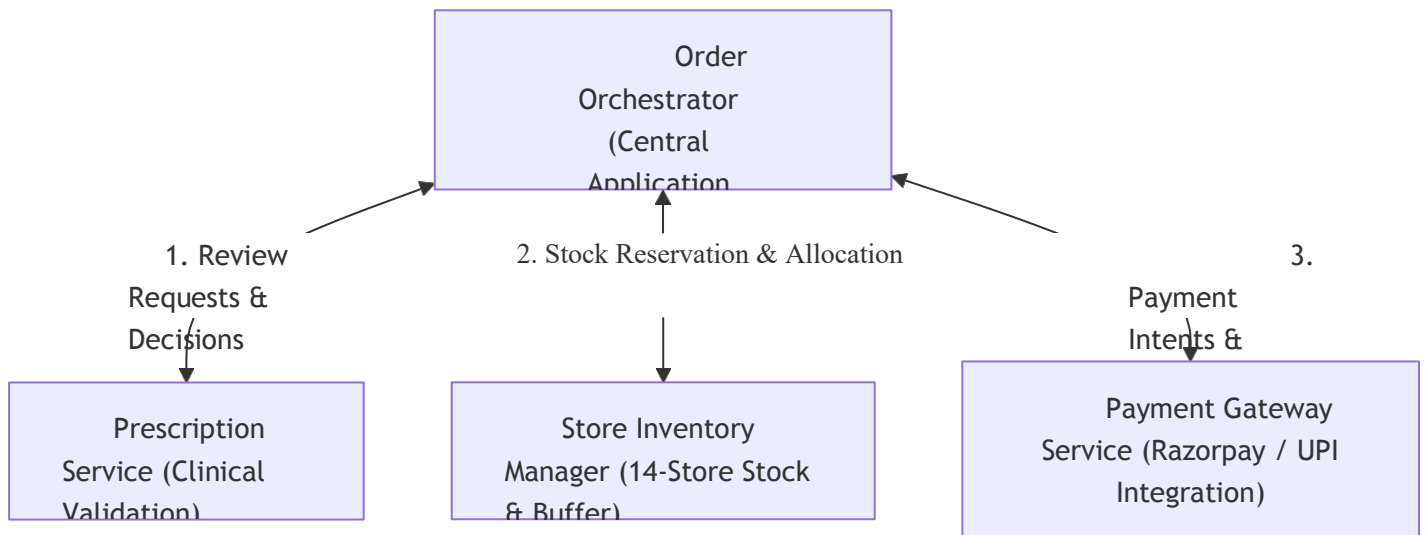
7.2 High Cohesion: Module-by-Module Evaluation

High cohesion ensures that every module in PharmaCart possesses a single, tightly defined responsibility (Single Responsibility Principle - SRP), minimizing side effects.

Module / Package	Cohesion Type	Primary Cohesive Responsibility	Non-Responsibilities (Delegated Elsewhere)
PrescriptionManagement	Functional Cohesion	Secure file ingestion, virus scanning, AES-256 vault storage, doctor credential parsing.	Does not calculate pricing, does not reserve stock, does not process payments.
PharmacistReviewPortal	Functional Cohesion	Renders dual-pane clinical viewer, enforces legal review checklists, captures digital signatures.	Does not interact with delivery couriers, does not modify store cash registers.
InventoryRoutingEngine	Informational Cohesion	Ingests 14-store POS delta feeds, computes effective available stock with safety buffer, selects optimal branch.	Does not validate prescription legality, does not handle customer authentication.
StoreCounterTerminal	Sequential Cohesion	Accepts picking tickets, guides store clerk shelf picking, scans barcodes, validates customer OTP.	Does not calculate tax rates, does not communicate directly with payment gateways.
PaymentService	Communicational Cohesion	Initializes payment gateway intents, processes webhook callbacks, triggers automated refunds.	Does not maintain order line items, does not access customer medical records.

7.3 Architectural Pattern: Event-Driven Mediator Model

To prevent circular dependencies (Prescription <-> Order <-> Inventory), PharmaCart employs the **Mediator Architectural Pattern** implemented via an Application Orchestrator (OrderOrchestrator).



- The PrescriptionService , StoreInventoryManager , and PaymentService never communicate directly with each other.
- All cross-boundary workflows are coordinated through the OrderOrchestrator , ensuring that changes to the 14-store POS integration protocol have zero ripple effects on the pharmacist review workflows.

8. Cross-Diagram Consistency Verification

To guarantee that the UML package is mathematically and structurally consistent across all artifacts, the following matrix verifies that all actors, classes, sequence lifelines, and states map directly to one another without orphaned or conflicting entities.

Entity / Concept	Use Case Diagram	Class Diagram	Sequence Diagram	State Machine Diagram
Customer	Present (Actor)	Customer Class	Present (Lifeline)	Trigger for DRAFT State
Prescription	Present (UC-02)	Prescription Class	Handled by RxSvc	Governs PENDING_RX_CHECK
Pharmacist Review	Present (UC-05)	PharmacistReview Class	Duty Pharmacist Actor	Triggers APPROVED / REJECTED
Store POS Inventory	Present (UC-06, 07)	StoreInventory Class	InvSvc & POS Lifelines	Governs STOCK_RESERVED
Store Allocation	Present (UC-08)	StoreOutlet Class	Router Lifeline	Enters ALLOCATED_TO_STORE
OTP Verification	Present (UC-1 0)	Method in Order	Final verification flow	Transitions to COMPLETED
Payment Gateway	Present (Supporting)	PaymentTransaction	PayGW Lifeline	Triggers CONFIRMED / FAILED

Formal Architecture Sign-Off

Designed & Submitted By:

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Lead Software Architect PharmaCart
Engineering Team Date: October 5, 2026

Reviewed & Approved By:

Head of Software Engineering PharmaCart
Project Delivery Team Date: October 5, 2026